WEEK 2 · LECTURE 2

Tokenization and Embeddings

How text becomes numbers a model can work with

BSc Data Science and Artificial Intelligence · Al-Zaytoonah University of Jordan · Dr. Akram Fatayer · 2026/2027

TODAY

By the end of this lecture you can…

01

Explain why models read subword tokens, not words or characters

02

Run byte-pair encoding by hand, and tell BPE, WordPiece and SentencePiece apart

03

Measure what a tokenizer costs in Arabic and in English

04

Tell a static embedding from a contextual one

LAST WEEK'S SURPRISE

Asked to continue "The capital of Jordan is", the model's top guess was not "Amman"

" Am" 37% of the probability went to a piece of a word

Today: what those pieces are, who chooses them, and what they cost in Arabic and English.

CHOOSING THE UNIT

Subword tokens are a compromise between characters and words

Unit	“unbelievable” becomes	Vocabulary	Sequence	Unseen words
Characters	u n b e l i e v a b l e	A few hundred	Very long	No problem
Words	unbelievable	Millions, never complete	Short	Unknown token
Subwords	un · believ · able	30,000–250,000, learned from data	In between	Split into known pieces

Arabic packs more into one word: وسيكتبونها, “and they will write it”, holds five pieces: و (and) + س (will) + يكتب (write) + ون (they) + ها (it).

BYTE-PAIR ENCODING

BPE learns its vocabulary by merging the most frequent pair, again and again

STEP 1

Split

Every training word becomes characters, plus an end-of-word mark _

STEP 2

Count

Every adjacent pair of symbols, weighted by how often its word occurs

STEP 3

Merge

The most frequent pair becomes one new symbol in the vocabulary

STEP 4

Repeat

Until the vocabulary reaches the size you chose

The list of merges, in order, **is** the tokenizer. Applying it to new text gives that text’s tokens.

BPE BY HAND

Three merges on the textbook example

Training words

Count	Symbols
5	l o w _
2	l o w e s t _
6	n e w e r _
3	w i d e r _
2	n e w _

First three merges

Step	Pair	Count	New symbol
1	e + r	9	er
2	er + _	9	er_
3	n + e	8	ne

e + r occurs 6 times in newer and 3 in wider: 9. Ties go to the pair found first.

APPLYING THE MERGES

New words never fail: they split into pieces the tokenizer knows

After six merges (er, er_, ne, new, lo, low), three words the tokenizer never saw:

lower

`low` `er_`

slower

`s` `low` `er_`

newest

`new` `e` `s` `t` `_`

Byte-level BPE (GPT-2, Qwen) starts from the 256 possible bytes instead of characters. Any text in any script can be encoded, and no token is ever “unknown”.

THREE ALGORITHMS

Same goal, a fixed vocabulary of frequent pieces, built three ways

Algorithm	How it builds the vocabulary	Marks word pieces as	Used by
BPE	Merges the most frequent pair, bottom up	A leading space: “ students”	GPT-2, Llama, Qwen
WordPiece	Merges the pair that most raises the likelihood of the training text	## on continuing pieces: “##bid”	BERT, mBERT, AraBERT
SentencePiece (unigram)	Starts with a large vocabulary and removes the pieces that matter least; treats the space as a symbol	_ at a word’s start: “_students”	XLM-R, T5, mT5

ONE SENTENCE, FIVE TOKENIZERS

Same meaning, very different counts

“The students travel from Irbid to Aqaba.”

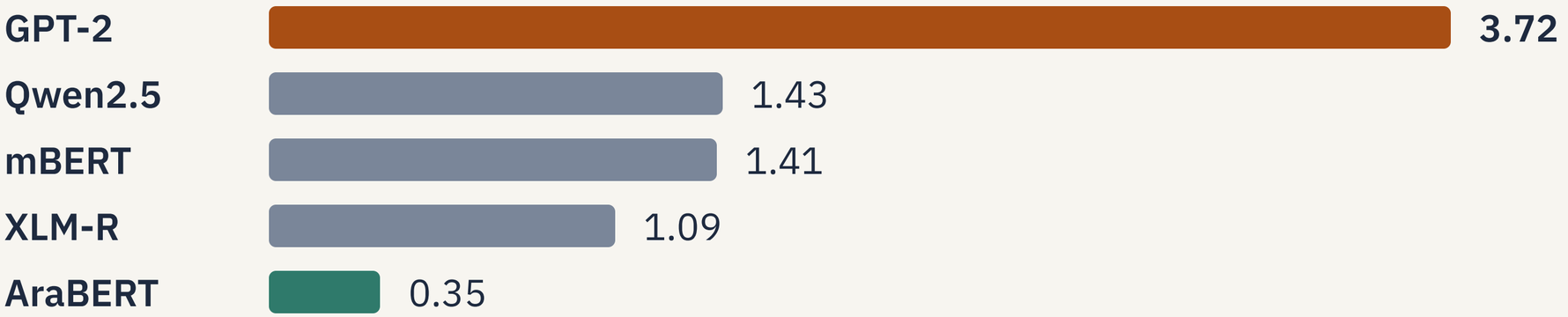
«يسافر الطلاب من إربد إلى العقبة.»

Tokenizer	Algorithm and training text	English tokens	Arabic tokens
GPT-2	Byte-level BPE, English web text	11	31
Qwen2.5	Byte-level BPE, multilingual	12	13
mBERT	WordPiece, 104 languages	11	15
XLM-R	SentencePiece, 100 languages	10	11
AraBERT	WordPiece, Arabic text	20	8

MEASURED ON 20 SENTENCE PAIRS

To say the same thing in Arabic, GPT-2 needs 3.7 times the tokens

Arabic tokens ÷ English tokens, for the same 20 sentences (1.0 = equal cost)



Fertility, tokens per word, tells the same story: 1.17 for English with GPT-2, 6.04 for Arabic. AraBERT is the mirror image: 1.27 for Arabic, 2.64 for English.

WHY THE GAP

Arabic is not harder; it is under-represented in the vocabulary

Training text

Merges go to frequent text. With little Arabic in it, Arabic words fall back to bytes, and each Arabic letter is two bytes in UTF-8.

Rich morphology

One root gives many surface words, and each form needs its own merges:

كتب · كاتب · مكتب · مكتبة · كتاب

Variation

Dialects, optional diacritics and spelling variants split the counts, so fewer forms become frequent enough to merge.

The fix is in the data: XLM-R, Qwen and Arabic-focused models give Arabic a fair share of the vocabulary.

WHAT THE GAP COSTS

Every extra token costs time, money and room in the context window

3.7×

GPU time, or price per request, for the same Arabic text with GPT-2’s tokenizer

678

Arabic words that fit in a 4,096-token context with GPT-2, against 3,501 English words

2,122

Arabic words in the same window with XLM-R’s multilingual tokenizer

Generation is one token per forward pass, so an answer that needs more tokens also takes longer to write.

FROM ID TO VECTOR

Each token ID selects one row of a learned table

Token
“ students”

→

ID
Its position in the vocabulary: one integer

→

Embedding
That row of the table: a vector of numbers, learned during training

151,936 × 1,536

Qwen2.5-1.5B’s table: 233 million numbers, about 15% of the whole model

250,037 × 384

The multilingual embedding model in today’s lab

WHAT THE VECTORS LEARN

Words used in similar places end up with similar vectors

> *“You shall know a word by the company it keeps.”*
>
> J. R. Firth, 1957

Counted from 20 sentences (Module 1)

Word pair	Cosine
brother · sister	1.00
amman · irbid	0.82
amman · students	0.31

word2vec (Mikolov et al. 2013) learned such vectors with a small network; an LLM learns its embedding table the same way, as part of training.

STATIC AND CONTEXTUAL

Only the layers above the table can tell two meanings apart

Word	Two uses compared	Static	Contextual
bank	river bank vs. salary at the bank	1.00	0.58
bank	salary at the bank vs. bank approved my loan	1.00	0.84
العين	spring in the valley vs. injured eye	1.00	0.90
العين	injured eye vs. eye examined by a doctor	1.00	0.94

The lab’s small model separates the two meanings of “bank” clearly, but the two meanings of العين only slightly.

ACROSS LANGUAGES

A multilingual model puts a sentence and its translation side by side

20 / 20

MSA sentences whose nearest English sentence is their own translation

4 / 5

Jordanian dialect sentences matched correctly

The miss: «يسلموا إيديك على مساعدتك بالتجربة»

“May your hands be safe” means “thank you”. The model matched the words, not the idiom, and chose “Please submit the assignment…”

WHAT COMES NEXT

This week: measure it yourself. Next week: attention

This week’s lab

Five tokenizers on Arabic and English, fertility and cost, static and contextual embeddings

Mini-assignment 1

Tokenizer audit. Individual, graded out of 10, due at the start of next week’s lab. Brief on the course page.

Reading

Before next week’s lecture on attention, read *Hands-On LLMs*, chapter 3.

Exit question: why can the same sentence need three times as many tokens with one tokenizer as with another?